PRACTICAL DBMS

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COURSE CODE: CSA0503

COURSE NAME: DATABASE MANAGEMENT SYSTEMS

EXP - 8 Query with Sub Query & Correlated Query

AIM:

To perform subquery and correlated query on the given relations.

DESCRIPTION:

SUBQUERY

A MySQL subquery is a query nested within another query such as SELECT, INSERT, UPDATE or DELETE. In addition, a MySQL subquery can be nested inside another subquery.

A MySQL subquery is called an inner query while the query that contains the subquery is called an outer query. A subquery can be used anywhere that expression is used and must be closed in parentheses.

CORRELATED QUERY:

A correlated subquery is a subquery that uses the data from the outer query. In other words, a correlated subquery depends on the outer query. A correlated subquery is evaluated once for each row in the outer query.

SYNTAX:

SUBQUERY:

SELECT c1, c2,...,cn FROM table

WHERE c1 IN (SELECTc1, c2,...,cn FROM table

WHERE where_conditions);

CORRELATED QUERY:

SELECT * FROM table_name

WHERE EXISTS (subquery);

Questions: Sub-Query and Correlated Sub-Query:

1. Which of the student's score is greater than the average score?

2. Which of the students' have written more than one assessment test?

3. Which faculty has joined recently and when?

4. List the course and score of assessments that have the value more than the average score each Course.

```
mysql> create table student(studentID int Primary key,studentName varchar(30));
Query OK, 0 rows affected (0.17 sec)

mysql> insert into student values(101,'Rahul'),(102,'Priya'),(103,'Arjun'),(104,'Sneha'),(105,'Rohan');
Query OK, 5 rows affected (0.07 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> select studentName,score from student join assessment on student.studentID=assessment.studentID where score>(select avg(score) from assessment);
+-------------+-------+
| studentName | score |
+-------------+-------+
| Rahul       |    85 |
| Rahul       |    90 |
| Arjun       |    95 |
+-------------+-------+
3 rows in set (0.04 sec)
```

```
mysql> select studentName from student where studentID in(select studentID from assessment group by studentID having count(*)>1);
+-------------+
| studentName |
+-------------+
| Rahul       |
| Arjun       |
+-------------+
2 rows in set (0.02 sec)

mysql> select facultyName,JoinDate from faculty where JoinDate=(select max(JoinDate) from faculty);
+-------------+------------+
| facultyName | JoinDate   |
+-------------+------------+
| Dr.Meena    | 2024-02-15 |
+-------------+------------+
1 row in set (0.01 sec)

mysql> select A.course,A.score from assessment A where A.score>(select avg(B.score) from assessment B where A.course=B.course);
+--------+-------+
| course | score |
+--------+-------+
| DBMS   |    85 |
| python |    90 |
| python |    95 |
| DBMS   |    80 |
+--------+-------+
4 rows in set (0.05 sec)
```